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Bereken de tweede afgeleide van f

$$f(x) = \sin(x)\cos(x)$$

$$f'(x) = \frac{d\sin(x)}{dx}\cos(x) + \frac{d\cos(x)}{dx}\sin(x) = \cos^2(x) - \sin^2(x)$$

$$f''(x) = \frac{d(\cos^2(x) - \sin^2(x))}{dx} = \frac{d(\cos^2(x))}{dx} - \frac{d(\sin^2(x))}{dx} =$$

$$-2\cos(x)\sin(x) - 2\sin(x)\cos(x) = -4\cos(x)\sin(x) = -2\sin(2x)$$

References